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VigilantAI — Business Requirements Document

> **Enterprise Security Intelligence Platform** Business Requirements Document — Version 1.0

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1.2 1. Document Control

Field	Value
Document Title	Business Requirements Document
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1.3 2. Revision History

Version	Date	Author	Changes
0.1	2026-07-21	Product Team	Initial draft — all sections
1.0	2026-07-21	Product Team	First release — pending stakeholder review

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1.4 3. Purpose

1.4.1 3.1 Document Objectives

This Business Requirements Document defines the business rationale, functional scope, and acceptance criteria for the VigilantAI Enterprise Security Intelligence Platform. It articulates the business problems that necessitate the platform, the operational goals it must achieve, and the measurable outcomes against which its success will be evaluated.

This document serves as the authoritative reference for business stakeholders, product managers, and engineering leadership to align on what VigilantAI must deliver, why it must deliver it, and how success will be determined.

1.4.2 3.2 Relationship to Executive Summary

This document is the natural extension of the Executive Summary (Document 01). Where the Executive Summary defines the product vision, positioning, architecture, and technology stack at a strategic level, this document translates those strategic directives into actionable business requirements with traceable acceptance criteria.

All terminology, stakeholder definitions, module references, business goals, and product scope described herein are consistent with and derived from the Executive Summary. In the event of ambiguity, the Executive Summary takes precedence.

1.4.3 3.3 Intended Audience

Role	Use of This Document
CEO / CIO	Strategic alignment, investment justification
CTO	Technical feasibility validation, resource planning
Product Managers	Requirements baseline, scope management, prioritization
Security Directors	Operational requirements validation, adoption planning
Operations Managers	Workflow alignment, productivity targets
Compliance Officers	Regulatory requirement mapping, audit readiness
Engineering Leadership	Acceptance criteria definition, implementation guidance

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1.5 4. Business Background

1.5.1 4.1 Enterprise Security Landscape

The global enterprise security market is undergoing a fundamental transformation. Organizations across corporate, industrial, healthcare, government, and critical infrastructure sectors operate camera fleets that generate thousands of hours of video per day. Yet the intelligence extracted from this footage remains minimal — less than 5% of captured video is ever reviewed by a human operator.

Legacy Video Management Systems (VMS) were designed for recording and storage. They were not built for real-time analysis, event correlation, or automated incident response. As camera fleets scale from dozens to thousands of cameras, the gap between data capture and operational intelligence widens.

1.5.2 4.2 Growth of AI-Powered Surveillance

Advances in computer vision, object detection, and deep learning have reached production-grade maturity for physical security applications. Models such as YOLO can detect, classify, and track objects —

persons, vehicles, restricted-zone intrusions — in real time from live camera feeds. The convergence of AI maturity, scalable cloud infrastructure, and declining GPU costs has made AI-powered surveillance technically and economically viable for the first time.

1.5.3 4.3 Enterprise Security Challenges

Despite increased investment in camera hardware, enterprises face persistent operational challenges:

- **Manual monitoring at scale is unsustainable.** A 200-camera facility produces over 4,800 hours of video per day. No Security Operations Center (SOC) can continuously review this volume. Operator fatigue sets in within minutes, and critical events are missed in real time.
- **Alert fatigue degrades response quality.** Motion-based and pixel-change detection generates enormous volumes of false alarms — shadows, weather effects, animals, lighting changes. Security teams learn to ignore alerts, creating a culture of desensitization that delays response to genuine threats.
- **Incident workflows are fragmented.** When an event is detected, operators must manually export footage, create incident reports, notify stakeholders, and track resolution across disconnected tools. This process is slow, error-prone, and creates gaps in the chain of custody for evidence.
- **Compliance requirements are escalating.** Regulations including GDPR, CCPA, HIPAA, and industry-specific standards impose strict requirements on video data handling, retention, access control, and audit trails. Legacy systems lack the granularity required to demonstrate compliance.

1.5.4 4.4 Digital Transformation in Physical Security

Physical security is following the same trajectory as cybersecurity — from reactive, hardware-centric models to proactive, software-defined, intelligence-driven operations. Organizations expect the same real-time visibility, automated response, and audit completeness from physical security that they have come to expect from their cyber defense platforms. This expectation creates both demand and urgency for platforms that can deliver intelligent security operations.

1.5.5 4.5 Industry Evolution

The competitive landscape spans legacy VMS vendors, cloud camera platforms, and AI analytics providers — but no existing solution combines real-time AI detection, configurable event processing, integrated incident management, evidence management with chain-of-custody tracking, and a unified Security Operations dashboard in a single platform. VigilantAI exists to fill this gap.

1.6 5. Business Problem Statement

1.6.1 5.1 Overview

Enterprise security operations face a convergence of challenges that legacy surveillance systems are structurally unable to address. These challenges are not isolated — they compound each other, creating systemic operational risk.

1.6.2 5.2 Detailed Problem Analysis

1.6.2.1 5.2.1 Manual Camera Monitoring at Scale A facility with 200 cameras produces over 4,800 hours of video per day. No security operations center can continuously monitor this volume. Operators experience fatigue within minutes, and critical events are missed in real time. As camera fleets grow — driven by expansion, new facilities, or regulatory requirements — the monitoring burden increases linearly while staffing budgets remain flat or decline.

1.6.2.2 5.2.2 Alert Fatigue and False Positives Legacy motion-based and pixel-change alerting generates enormous volumes of false alarms. Studies indicate that 80–95% of alerts from conventional systems are false positives — triggered by shadows, weather conditions, animals, lighting changes, or non-threatening human activity. Security teams learn to ignore alerts, creating a culture of desensitization that delays response to genuine threats. This is not a training problem — it is a systems design problem.

1.6.2.3 5.2.3 Slow and Error-Prone Investigations When an incident occurs, operators must manually identify relevant footage, export video clips, assemble incident reports, and track evidence through disconnected tools. This process takes hours when it should take minutes. The fragmented workflow creates gaps in the chain of custody, undermining the evidentiary value of footage for legal, regulatory, or internal proceedings.

1.6.2.4 5.2.4 Evidence Management Issues Evidence storage, access control, retention, and audit trails are critical requirements for enterprise security. Legacy systems provide no automated chain-of-custody tracking, no integrity verification, and no role-based access enforcement on evidence. Manual evidence management is labor-intensive, error-prone, and fails under regulatory scrutiny.

1.6.2.5 5.2.5 Lack of Correlated Intelligence Individual camera events exist in isolation. Without cross-camera correlation, pattern analysis, and contextual enrichment, security teams cannot identify coordinated threats, recurring patterns, or systemic vulnerabilities. A person crossing three cameras in sequence appears as three independent events, not a single coordinated movement.

1.6.2.6 5.2.6 Compliance Complexity Regulations including GDPR (video data handling, retention, access control), CCPA (data access and deletion rights), HIPAA (healthcare facility security), SOC 2 (audit trails, change management), and IEC 62443 (industrial cybersecurity) impose increasingly strict requirements. Legacy surveillance systems lack the granularity needed to demonstrate compliance during audits.

1.6.2.7 5.2.7 Growing Camera Fleets and Operational Inefficiency As camera fleets scale from dozens to thousands of cameras, the operational burden grows linearly — more cameras to monitor, more footage to review, more alerts to triage, more incidents to manage. Legacy systems offer no automation or intelligence to offset this scaling penalty. The result is declining security coverage per dollar invested.

1.6.2.8 5.2.8 High Staffing Cost Security operations are labor-intensive. Staffing a 24/7 monitoring operation for a 500-camera facility requires multiple shifts of trained operators. Labor costs represent 60–80% of physical security operating budgets. Without automation, cost reduction is impossible.

1.6.2.9 5.2.9 Limited Visibility Executives and security leadership lack real-time visibility into physical security posture. Status reports are manual, retrospective, and incomplete. There is no equivalent of the dashboards and metrics that cyber security teams use to understand their operational posture in real time.

1.6.3 5.3 Business Impact

Problem Area	Business Impact
Manual monitoring	Missed incidents, delayed response, increased liability
False positives	Operator desensitization, wasted labor, eroded trust in the system
Slow investigations	Extended time-to-resolution, increased operational cost
Evidence management gaps	Legal exposure, compliance violations, invalidated evidence
Lack of correlation	Inability to detect coordinated threats, systemic vulnerabilities missed
Compliance gaps	Regulatory fines, audit failures, reputational damage
Operational inefficiency	Declining security coverage per dollar invested
High staffing costs	Unsustainable cost trajectory as camera fleets grow
Limited visibility	Strategic decision-making without current security posture data

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1.7 6. Business Opportunity

1.7.1 6.1 Market Opportunity

The global video surveillance market is projected to exceed $80 billion by 2028. However, the intelligence layer on top of surveillance infrastructure remains significantly underpenetrated. Enterprises spend billions on camera hardware but capture less than 5% of the available security intelligence from their video feeds.

Three converging forces create a distinct market opportunity:

1. **AI maturity** — Computer vision models have reached production-grade accuracy for physical security use cases.
2. **Cloud infrastructure** — Scalable compute and storage eliminate the cost barriers that previously constrained on-premise AI deployment.
3. **Security operations burden** — Rising labor costs and staffing shortages make automation an operational necessity, not a luxury.

VigilantAI is positioned to capture the emerging **AI-powered Physical Security Intelligence** market segment — a category distinct from both legacy VMS and cloud-managed camera platforms.

1.7.2 6.2 Industry Demand

Demand Driver	Evidence
Rising security staffing costs	60–80% of security budgets allocated to labor
Regulatory pressure	GDPR, CCPA, HIPAA compliance requirements expanding

Demand Driver	Evidence
Camera fleet growth	Enterprise camera fleets growing 15–25% annually
AI technology readiness	Computer vision models achieving >95% accuracy on core tasks
Digital transformation in security	Convergence of physical and cyber security operations
Post-pandemic security requirements	Increased demand for remote monitoring and automated response

1.7.3 6.3 Operational Improvements

Current State	Future State with VigilantAI
Manual monitoring of live feeds	AI-driven detection with operator escalation only for genuine events
80–95% false positive rate	AI-classified alerts with <10% false positive rate
Hours to assemble evidence	Seconds to retrieve correlated evidence packages
Disconnected incident tracking	Unified incident lifecycle from detection through resolution
No cross-camera intelligence	Correlated event timelines across camera fleet
Manual compliance reporting	Automated audit trails and compliance dashboards
No executive security visibility	Real-time KPI dashboards for security leadership

1.7.4 6.4 Business Value

- **Reduced operational cost** — Automation reduces the number of operators required per facility.
- **Improved detection accuracy** — AI classification significantly reduces false-positive rates.
- **Faster investigations** — Correlated event timelines and evidence packaging accelerate post-incident analysis.
- **Compliance readiness** — Automated audit trails and evidence management support regulatory requirements.
- **Unified security view** — Single platform consolidates monitoring, incident management, and reporting.

1.7.5 6.5 Return on Investment (ROI)

ROI Component	Expected Impact
Operator headcount reduction	30–50% reduction in monitoring staff per facility
Investigation time reduction	40–60% faster time-to-resolution
False positive reduction	80–90% fewer alerts requiring human triage
Compliance cost avoidance	Elimination of manual audit preparation labor
Incident cost reduction	Earlier detection reduces damage and liability exposure

1.7.6 6.6 Competitive Advantage

No existing solution combines real-time AI detection, configurable event processing, integrated incident management, evidence management with chain-of-custody integrity, and a unified Security Operations dashboard in a single platform. Legacy vendors bolt AI onto VMS. Cloud platforms manage cameras but not intelligence. Analytics providers detect events but do not manage incidents. VigilantAI fills the gap between detection and response.

1.7.7 6.7 Customer Value

Customer Segment	Primary Value Delivered
Corporate Offices	After-hours intrusion detection, tailgating alerts, restricted area enforcement
Manufacturing	PPE compliance, safety zone enforcement, equipment theft prevention
Healthcare	Patient area security, pharmacy access monitoring, infant protection
Government	Compliance-driven security, classified area protection
Critical Infrastructure	Perimeter defense, sabotage prevention, regulatory compliance

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1.8 7. Vision Statement

1.8.1 7.1 Business Vision

VigilantAI exists to close the gap between passive video recording and active security intelligence. Most enterprise surveillance systems generate thousands of hours of footage that are never reviewed. Security teams are overwhelmed by alert fatigue, manual workflows, and the inability to correlate events across a distributed camera fleet. The result is delayed response times, missed incidents, and expensive post-incident forensic investigations.

1.8.2 7.2 Mission Alignment

The mission of VigilantAI is to provide enterprises with an intelligent security platform that automates threat detection, streamlines incident response, and delivers operational visibility across the entire camera fleet — reducing risk while lowering the operational burden on security teams.

1.8.3 7.3 Long-Term Direction

VigilantAI shifts the paradigm from *record and review* to *detect and respond*. By applying computer vision at the edge of the camera stream and orchestrating events through a centralized intelligence layer, the platform enables security teams to operate proactively rather than reactively.

1.8.4 7.4 Future Growth

The long-term vision positions VigilantAI as the intelligent fabric connecting physical security infrastructure with enterprise security operations — a platform where every camera becomes a sensor, every event becomes actionable intelligence, and every incident is tracked from detection through resolution.

1.9 8. Business Goals

1.9.1 8.1 Year 1 Goals

Goal #	Goal	Success Measure
G-01	Deploy production instances across 3–5 enterprise pilot customers	≥ 5 active customer deployments
G-02	Achieve measurable reduction in mean-time-to-detect (MTTD) for security incidents	MTTD < 5 seconds
G-03	Establish the platform as a viable alternative to legacy VMS for intelligent monitoring	Positive customer evaluations and renewals

1.9.2 8.2 Year 2 Goals

Goal #	Goal	Success Measure
G-04	Scale to 20+ enterprise deployments across multiple verticals	≥ 20 active deployments
G-05	Expand AI model coverage to support additional detection scenarios	≥ 10 supported detection types
G-06	Build integration ecosystem with access control, HR, and SIEM platforms	≥ 5 third-party integrations available

1.9.3 8.3 Year 3 Goals

Goal #	Goal	Success Measure
G-07	Achieve market recognition as a category leader in AI-driven physical security	Analyst recognition, press coverage
G-08	Support 10,000+ camera deployments per customer instance	Demonstrated at ≥ 3 customer sites
G-09	Generate recurring revenue through SaaS and managed service offerings	Revenue target achieved

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1.10 9. Business Objectives

1.10.1 9.1 SMART Objectives

Obj #	Objective	Specific	Measurable	Achievable	Relevant	Time-Bound
OBJ-01	Reduce mean time to detect security incidents	AI-powered real-time detection	MTTD reduced to < 5 seconds from baseline	Yes	Core product value proposition	Phase 1
OBJ-02	Reduce false-positive alert rate	AI classification of detection events	False-positive rate reduced to < 10%	Yes	Directly impacts operator trust	Phase 1
OBJ-03	Automate incident lifecycle management	Detection-to-resolution workflow	40% reduction in mean time to resolve	Yes	Operational efficiency	Phase 1
OBJ-04	Eliminate manual evidence assembly	Automated evidence clip creation	Evidence retrieval time < 10 seconds	Yes	Compliance and investigation	Phase 1
OBJ-05	Deliver real-time operational visibility	Security Operations Dashboard	Dashboard load time < 2 seconds	Yes	Executive and operational needs	Phase 1
OBJ-06	Support enterprise camera fleet at scale	Scalable camera ingestion	50 to 10,000+ cameras per deployment	Yes	Enterprise customer requirement	Phase 1
OBJ-07	Deploy pilot customers	Production deployments	≥ 5 enterprise customers within 12 months	Yes	Market validation	Year 1
OBJ-08	Expand detection scenarios	Additional AI model coverage	≥ 10 detection types within 18 months	Yes	Market differentiation	Year 1–2
OBJ-09	Build integration ecosystem	Third-party integrations	≥ 5 integrations (access control, SIEM, HR)	Yes	Enterprise adoption requirement	Year 2
OBJ-10	Achieve SaaS readiness	Multi-tenant cloud platform	SaaS offering operational within 24 months	Yes	Recurring revenue model	Year 2

1.11 10. Stakeholders

1.11.1 10.1 Stakeholder Matrix

Stakeholder	Role	Responsibilities	Influence	Interest	Success Criteria
CEO	Executive Sponsor	Strategic direction, funding, organizational alignment	High	High	ROI achieved, market position established

Stakeholder	Role	Responsibilities	Influence	Interest	Success Criteria
CTO	Technical Executive Sponsor	Technology strategy, architecture governance, build-vs-buy	High	High	Platform meets performance and scalability targets
CIO	IT Executive Sponsor	Infrastructure, deployment, integration with enterprise IT	High	Medium	Deployment fits existing IT operations model
Product Management	Product Owner	Requirements, prioritization, roadmap, market fit	High	High	Product delivers business value, customers adopt
Engineering Leadership	Technical Delivery	Architecture, implementation, quality, delivery timelines	High	High	Platform built to spec, within budget and timeline
Security Operations	Primary End User	Platform operation, alert triage, incident management	Medium	High	Platform reduces workload, improves response quality
Security Director	Operational Sponsor	Security strategy, team management, budget	High	High	Security posture measurably improved
Compliance Officer	Regulatory Stakeholder	Regulatory alignment, audit preparation, policy enforcement	Medium	High	Compliance requirements demonstrably met
Operations Manager	Workflow Owner	Operational process design, team productivity	Medium	High	Operational efficiency targets achieved
IT Administrator	Technical Operator	Platform deployment, configuration, maintenance	Medium	Medium	Platform operable with existing IT skills
Incident Investigator	Specialized End User	Post-incident forensics, evidence assembly, reporting	Low	High	Investigation time significantly reduced
Executive Leadership	Strategic Audience	Security posture visibility, strategic risk assessment	High	Low	Real-time visibility into security operations
Legal Counsel	Compliance Stakeholder	Legal risk assessment, evidence admissibility, data handling	Medium	Medium	Evidence and audit trails meet legal standards
Facilities Manager	Cross-functional Stakeholder	Physical security integration with building operations	Low	Medium	Security operations aligned with facility operations
QA / Test Engineering	Quality Gatekeeper	Platform validation, defect prevention, acceptance testing	Medium	High	Platform meets quality and acceptance criteria
DevOps / SRE	Deployment and Operations	Deployment, monitoring, uptime, scalability	Medium	High	99.95% uptime achieved

1.12 11. Business Scope

1.12.1 11.1 In Scope (MVP)

Scope Item	Description
Live camera stream ingestion via RTSP	Connect to IP camera fleets using industry-standard RTSP protocol
AI object detection	Detect persons, vehicles, and objects of interest in live video feeds
Restricted zone monitoring	Detect unauthorized presence in configured restricted zones
Real-time event generation and classification	Generate classified security events from AI detections
Incident creation and lifecycle management	Create, assign, track, and resolve security incidents
Evidence storage with chain-of-custody tracking	Preserve forensic evidence with timestamping, access tracking, and integrity verification
Configurable rule engine	Business rules governing event evaluation, filtering, escalation, and routing
Security Operations Dashboard	Real-time console for monitoring, alert management, and incident workflows
Event timeline with filtering and search	Temporal view of all events with filtering by camera, type, severity
Camera fleet management and health monitoring	Centralized camera registration, health monitoring, and fleet visibility
Role-based access control	Role-based permissions across all platform modules
Audit logging	Immutable audit trail for all user actions and system events
REST API and WebSocket streaming	API access for integration and real-time data streaming
Docker-based deployment	Containerized deployment for environment consistency

1.12.2 11.2 Out of Scope (MVP)

Scope Item	Reason
On-premise hardware appliance offering	Software-first approach; hardware in Phase 3
Custom model training interface	Deferred to Phase 2; pre-trained models in MVP
Video analytics marketplace / plugin architecture	Deferred to Phase 3
Mobile application	Deferred to Phase 2; web dashboard mobile-responsive
Multi-tenant SaaS control plane	Deferred to Phase 2
Video management features (playback, export, NVR)	Deferred to integration partners — not a VMS replacement
Edge AI deployment (on-camera inference)	Deferred to Phase 4
Face recognition	Deferred to Phase 4
License plate recognition	Deferred to Phase 4

Scope Item	Reason
Weapon detection	Deferred to Phase 4

1.12.3 11.3 Future Scope (Phase 2–5)

Scope Item	Target Phase
Advanced detection scenarios (intrusion patterns, loitering, crowd analysis)	Phase 2
Multi-camera event correlation	Phase 2
Custom rule builder UI	Phase 2
Alert escalation and notification channels	Phase 2
Reporting and analytics module	Phase 2
PostgreSQL production deployment support	Phase 2
Mobile-responsive dashboard	Phase 2
Access control system integration	Phase 3
SIEM platform integration	Phase 3
Multi-site management console	Phase 3
SSO / SAML authentication	Phase 3
High availability and failover	Phase 3
Face recognition with watchlist management	Phase 4
License plate recognition and vehicle tracking	Phase 4
Weapon detection and threat alerting	Phase 4
Fire and smoke detection	Phase 4
PPE compliance detection	Phase 4
Crowd density analytics	Phase 4
Predictive threat intelligence engine	Phase 5
Digital twin integration	Phase 5
Cloud SaaS platform for managed deployments	Phase 5

1.13 12. Current Business Process (AS-IS)

1.13.1 12.1 Current Surveillance Workflow

```
flowchart TB
    A[Camera Captures Footage] --> B[Video Sent to NVR / VMS]
    B --> C[Footage Stored on Disk]
    C --> D{Incident Reported?}
    D -->|Yes| E[Operator Manually Reviews Footage]
    D -->|No| F[Footage Aged and Deleted]
    E --> G[Operator Identifies Relevant Clip]
    G --> H[Operator Manually Exports Clip]
    H --> I[Operator Creates Incident Report]
```

```
    I --> J[Clip Shared via Email or File Share]
    J --> K[Incident Resolved Outside System]
```

1.13.2 12.2 Current Incident Workflow

```
flowchart TB
    A[Alert Generated — Motion / Pixel Change] --> B{Operator Notices Alert?}
    B -->|No — Missed| C[Event Lost — No Record]
    B -->|Yes| D[Operator Views Live Feed]
    D --> E{Genuine Threat?}
    E -->|No — False Positive| F[Operator Dismisses Alert]
    E -->|Yes| G[Operator Contacts Response Team]
    G --> H[Operator Manually Exports Footage]
    H --> I[Operator Creates Report in Separate System]
    I --> J[Incident Tracked in Email or Spreadsheet]
    J --> K[Resolution Not Captured in Surveillance System]
```

1.13.3 12.3 Current Evidence Workflow

```
flowchart TB
    A[Incident Identified] --> B[Operator Locates Relevant Camera Feed]
    B --> C[Operator Reviews Hours of Footage]
    C --> D[Operator Identifies Relevant Time Window]
    D --> E[Operator Exports Video Clip from NVR]
    E --> F[Clip Saved to Local Drive or USB]
    F --> G[Clip Attached to Email or Report]
    G --> H{Chain of Custody Tracked?}
    H -->|No — Most Cases| I[Evidence Integrity Unverified]
    H -->|Yes — Rare| J[Manual Log of Evidence Handling]
    I --> K[Evidence Submitted for Investigation]
    J --> K
```

1.13.4 12.4 Current Reporting Workflow

```
flowchart TB
    A[Security Manager Needs Report] --> B[Operator Compiles Data from Multiple Sources]
    B --> C[Manual Count of Incidents from Email / Spreadsheet]
    C --> D[Manual Review of Alert Logs]
    D --> E[Manual Assembly of Summary Report]
    E --> F[Report Delivered to Leadership]
    F --> G{Report Accurate and Current?}
    G -->|Often No| H[Data is Retrospective and Incomplete]
    G -->|Sometimes| I[Leadership Has Limited Visibility]
```

1.13.5 12.5 Key Pain Points in AS-IS State

Process Area	Pain Point
Monitoring	Manual review of live feeds; operator fatigue; critical events missed
Alerting	80–95% false positives; operators desensitized
Incident Management	Disconnected tracking across email, spreadsheets, and external tools
Evidence Management	Manual export; no chain of custody; no integrity verification
Reporting	Manual assembly; retrospective data; incomplete and error-prone
Cross-Camera Intelligence	None — each camera is an isolated data source
Compliance	No automated audit trails; manual compliance reporting

---

1.14 13. Future Business Process (TO-BE)

1.14.1 13.1 Future AI-Driven Monitoring Workflow

```
flowchart TB
    A[Camera Fleet — RTSP] --> B[VigilantAI Camera Gateway]
    B --> C[AI Detection Engine — Real-Time Analysis]
    C --> D[Object Detection — Persons, Vehicles, Objects]
    D --> E[Classification and Confidence Scoring]
    E --> F[Zone Evaluation — Restricted Area Check]
    F --> G{Rule Engine Match?}
    G -->|No Match| H[Event Logged — Info Level]
    G -->|Match| I[Security Event Generated]
    I --> J[Alert Created and Classified]
    J --> K[Incident Auto-Created if Threshold Met]
    K --> L[Evidence Clip Auto-Captured]
    L --> M[Operator Notified via Dashboard]
    M --> N[Operator Reviews — Acknowledges or Escalates]
    N --> O[Incident Tracked Through Resolution]
```

1.14.2 13.2 Automated Incident Handling Workflow

```
flowchart TB
    A[Security Event Detected] --> B[Rule Engine Evaluates Conditions]
    B --> C{Severity Classification}
    C -->|Critical| D[Immediate Alert — Auto-Escalate]
    C -->|High| E[Priority Alert — Assign to Operator]
    C -->|Medium| F[Standard Alert — Queue for Review]
    C -->|Low| G[Informational — Log Only]
    D --> H[Incident Created — SLA Timer Starts]
    E --> H
    F --> H
    H --> I[Operator Acknowledges]
    I --> J[Operator Reviews Evidence]
    J --> K[Operator Adds Investigation Notes]
    K --> L{Resolution}
```

```
    L -->|Resolved| M[Incident Closed — Resolution Documented]
    L -->|Escalated| N[Incident Escalated — New Owner Assigned]
    L -->|False Positive| O[Incident Closed — Marked as False Positive]
    M --> P[Audit Trail Recorded]
    N --> P
    O --> P
```

1.14.3 13.3 Real-Time Monitoring Dashboard Workflow

```
flowchart TB
    A[Camera Fleet — Live Feeds] --> B[AI Detection Engine]
    B --> C[Event Processor]
    C --> D[Real-Time Dashboard]
    D --> E[Live Camera Grid]
    D --> F[Active Alert Console]
    D --> G[Incident Queue]
    D --> H[Fleet Health Status]
    D --> I[KPI Metrics Panel]
    E --> J[Operator Selects Camera — Views Live Feed]
    F --> K[Operator Acknowledges Alert — Views Context]
    G --> L[Operator Manages Incident Lifecycle]
    H --> M[Admin Monitors Camera Health]
    I --> N[Leadership Reviews Security Posture]
```

1.14.4 13.4 Evidence Automation Workflow

```
flowchart TB
    A[Security Event Detected] --> B[Evidence Clip Created Automatically]
    B --> C[Content Hash Generated — SHA-256]
    C --> D[Evidence Stored with Timestamp and Metadata]
    D --> E[Access Control Applied — Role-Based]
    E --> F[Incident Associated with Evidence]
    F --> G[Operator Requests Evidence]
    G --> H{Authorization Check}
    H -->|Authorized| I[Access Logged — Evidence Served]
    H -->|Unauthorized| J[Access Denied — Logged]
    I --> K[Operator Views Evidence]
    K --> L{Export Required?}
    L -->|Yes| M[Export Authorization Required]
    L -->|No| N[Session Complete]
    M --> O{Export Approved?}
    O -->|Yes| P[Export Package Created — Logged]
    O -->|No| Q[Export Denied — Logged]
    P --> R[Chain of Custody Record Complete]
```

1.14.5 13.5 Key Improvements in TO-BE State

Process Area	Improvement
Monitoring	AI analyzes all cameras simultaneously; operators focus on genuine events
Alerting	AI-classified alerts with <10% false positive rate
Incident Management	Unified lifecycle from detection through resolution in single platform
Evidence Management	Automated capture, chain-of-custody tracking, integrity verification
Reporting	Real-time dashboards and automated compliance reports
Cross-Camera Intelligence	Correlated event timelines across camera fleet
Compliance	Automated audit trails, role-based access, retention policies

---

1.15 14. Business Capabilities

1.15.1 14.1 Business Capability Matrix

Capability ID	Capability	Description	Priority	Dependencies
CAP-01	Camera Monitoring	Real-time monitoring of live camera feeds with AI-powered analysis	Critical	Camera Gateway, AI Detection
CAP-02	Threat Detection	AI-based detection and classification of security threats in real time	Critical	AI Detection Engine
CAP-03	AI Analytics	Computer vision analytics including object detection, classification, tracking	Critical	AI Detection Engine
CAP-04	Event Processing	Real-time event generation, classification, and correlation	Critical	Event Processor, Rule Engine
CAP-05	Alert Management	Alert creation, classification, prioritization, and delivery	High	Event Processor, Alert Dispatcher
CAP-06	Incident Management	Incident lifecycle from creation through investigation, resolution, archival	High	Incident Manager, Evidence Manager
CAP-07	Evidence Management	Automated evidence capture, storage, chain-of-custody, and retrieval	High	Evidence Manager, Camera Gateway
CAP-08	Rule Management	Configurable business rules for event evaluation, filtering, escalation	High	Rule Engine
CAP-09	Security Operations Dashboard	Real-time operational visibility through unified console	High	All backend modules
CAP-10	Camera Fleet Management	Centralized fleet registration, health monitoring, and configuration	High	Camera Fleet Manager
CAP-11	Reporting	Operational, compliance, and strategic security reporting	Medium	Event Processor, Incident Manager
CAP-12	Compliance	Regulatory compliance support including audit trails, retention, access control	Medium	Audit Service, Authorization
CAP-13	User Administration	User management, role assignment, and permission enforcement	Medium	Authentication, Authorization

Capability ID	Capability	Description	Priority	Dependencies
CAP-14	Audit	Immutable audit trail for all user and system actions	Medium	Audit Service
CAP-15	Integration	API-based integration with external systems (SIEM, access control, HR)	Medium	API Gateway
CAP-16	Notification	Alert delivery via email, SMS, webhook, and dashboard	High	Alert Dispatcher

---

1.16 15. User Personas

1.16.1 15.1 Security Operator / Monitor

Attribute	Detail
Role	Security Operator / Monitor
Goals	Respond to genuine threats quickly; minimize false-positive burden; maintain situational awareness
Responsibilities	Monitor live feeds, acknowledge alerts, triage incidents, respond to security events, escalate as needed
Daily Activities	Review alert console, investigate flagged events, acknowledge or dismiss alerts, create incident notes, coordinate with response teams
Pain Points	Alert fatigue from false positives; too many tools to monitor; manual evidence assembly; no cross-camera visibility
Success Measures	Fast acknowledgment of genuine alerts; low false-positive dismissal rate; complete incident documentation

1.16.2 15.2 Security Manager / Operations Manager

Attribute	Detail
Role	Security Manager / Operations Manager
Goals	Ensure operational efficiency; meet security KPIs; manage team effectively
Responsibilities	Oversee security operations, manage operator workflows, review performance metrics, allocate resources, report to leadership
Daily Activities	Review KPI dashboard, monitor team performance, manage shift assignments, review open incidents, prepare reports for leadership
Pain Points	Lack of real-time visibility into operations; manual reporting; difficulty correlating events across sites; staffing challenges
Success Measures	Operational KPIs met; team productivity improved; security posture measurably enhanced

1.16.3 15.3 Security Director

Attribute	Detail
Role	Security Director
Goals	Set security strategy; ensure regulatory compliance; manage security budget; demonstrate ROI
Responsibilities	Define security policy, manage budget, review strategic risk metrics, liaise with executive leadership, ensure compliance
Daily Activities	Review executive dashboard, assess risk posture, approve policy changes, meet with executive leadership, evaluate vendor solutions
Pain Points	No real-time visibility into physical security posture; manual reporting is retrospective and incomplete; difficulty justifying security spend without data
Success Measures	Security posture improved; compliance requirements met; budget efficiency demonstrated

1.16.4 15.4 IT Administrator

Attribute	Detail
Role	IT / Systems Administrator
Goals	Deploy, configure, and maintain the platform; ensure system health and uptime
Responsibilities	Platform deployment, configuration, integration with existing IT infrastructure, monitoring, troubleshooting, maintenance
Daily Activities	Monitor system health, manage camera fleet configuration, handle user access requests, apply updates, troubleshoot issues
Pain Points	Limited documentation; compatibility issues with diverse camera vendors; complex deployment procedures; integration challenges with existing systems
Success Measures	Platform uptime meets SLA; deployment time minimized; integration issues resolved quickly

1.16.5 15.5 Compliance Officer

Attribute	Detail
Role	Compliance Officer
Goals	Ensure regulatory adherence; prepare for audits; manage data governance
Responsibilities	Review audit trails, generate compliance reports, ensure data handling meets regulatory requirements, prepare for external audits
Daily Activities	Review access logs, monitor evidence retention policies, generate compliance reports, address audit findings, update compliance documentation
Pain Points	Manual audit preparation is time-consuming; evidence chain-of-custody gaps; data retention policy enforcement is manual; regulatory requirements vary by geography

Attribute	Detail
Success Measures	Audit findings minimized; compliance reports generated on schedule; data governance requirements met

1.16.6 15.6 Incident Investigator

Attribute	Detail
Role	Incident Investigator
Goals	Conduct thorough post-incident investigations; assemble complete evidence packages; determine root cause
Responsibilities	Investigate security incidents, assemble evidence, reconstruct event timelines, produce investigation reports
Daily Activities	Review incident details, retrieve evidence clips, analyze event timelines, cross-reference camera feeds, compile investigation reports
Pain Points	Evidence scattered across systems; manual clip assembly; no chain-of-custody tracking; difficulty correlating events across multiple cameras and timeframes
Success Measures	Complete evidence packages assembled quickly; root cause identified; investigation reports comprehensive and accurate

1.16.7 15.7 Executive Leadership

Attribute	Detail
Role	Executive Leadership (CEO, CTO, CIO)
Goals	Understand security posture; assess risk exposure; evaluate ROI of security investments
Responsibilities	Review strategic risk metrics, approve security investments, assess organizational risk posture, make strategic decisions
Daily Activities	Review executive dashboard, assess key risk indicators, review incident trends, evaluate security spending effectiveness
Pain Points	Lack of real-time visibility; reporting is retrospective and manual; difficulty correlating physical security with business risk; no standardized metrics
Success Measures	Real-time visibility into security posture; clear ROI metrics; risk exposure understood and managed

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1.17 16. Business Requirements

1.17.1 16.1 Monitoring Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-001	The platform shall ingest live video streams from IP cameras via RTSP	Critical	Foundation for all AI-driven security capabilities	Platform connects to RTSP camera feeds and delivers frames for downstream processing
BR-002	The platform shall display live camera feeds in the Security Operations Dashboard	Critical	Operators require real-time visual monitoring capability	Live feeds rendered in dashboard within 2 seconds of selection
BR-003	The platform shall support concurrent monitoring of multiple camera feeds	Critical	Operators must monitor multiple cameras simultaneously	Dashboard supports minimum 4 simultaneous live feeds per operator session
BR-004	The platform shall monitor camera fleet health and report camera status	High	Degraded cameras reduce security coverage	Camera health status displayed in fleet management view; offline cameras flagged within 60 seconds
BR-005	The platform shall support camera fleet organization by site, building, and zone	High	Enterprise deployments require hierarchical camera organization	Cameras organized hierarchically; site/building/zone structure configurable

1.17.2 16.2 Alert Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-006	The platform shall generate real-time alerts when AI detection meets rule criteria	Critical	Core value proposition — automated threat detection	Alerts generated within 5 seconds of detection event
BR-007	The platform shall classify alerts by severity (Critical, High, Medium, Low)	High	Operators must prioritize response based on threat severity	Every alert carries a severity classification; severity is configurable via rules
BR-008	The platform shall reduce false-positive alerts through AI-based classification	Critical	Alert fatigue directly impacts operator response quality	False-positive rate < 10% of total alerts after AI classification
BR-009	The platform shall deliver alerts to the Security Operations Dashboard in real time	Critical	Operators must see alerts immediately for rapid response	Alerts rendered in dashboard console within 5 seconds of generation
BR-010	The platform shall support alert acknowledgment by authorized operators	High	Every alert requires human review and acknowledgment	Operators can acknowledge alerts; acknowledgment timestamp recorded

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-011	The platform shall provide alert filtering by camera, zone, severity, and time range	High	Operators must focus on relevant alerts efficiently	Filter controls available on alert console; filters applied within 1 second
BR-012	The platform shall support alert escalation workflows	High	Critical alerts require immediate escalation to supervisors	Escalation rules configurable; escalation triggers notification to designated roles

1.17.3 16.3 Incident Management Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-013	The platform shall create incidents automatically from correlated security events	Critical	Automated incident creation reduces manual workflow burden	Incidents created within 10 seconds of event correlation threshold being met
BR-014	The platform shall support manual incident creation by authorized operators	High	Operators must be able to create incidents from observed activity	Manual incident creation form available; required fields enforced
BR-015	The platform shall assign incidents to designated operators	High	Every incident must have an owner for accountability	Assignment functionality available; assignment recorded with timestamp
BR-016	The platform shall track incident status through its complete lifecycle	Critical	Incidents must be tracked from creation through resolution	Status transitions (Open → Acknowledged → Investigating → Resolved/Closed) tracked
BR-017	The platform shall enforce SLA timers on incident response and resolution	High	Delayed response increases risk exposure	SLA timers configurable per severity; SLA breaches flagged to management
BR-018	The platform shall support investigation notes on incidents	High	Investigators must document findings and observations	Notes attached to incidents with author, timestamp, and content
BR-019	The platform shall associate evidence clips with incidents	Critical	Evidence linking is essential for investigation and prosecution	Evidence clips linked to incidents; linked evidence visible in incident detail
BR-020	The platform shall provide incident search and filtering	High	Operators must locate incidents efficiently across large volumes	Search by incident ID, status, severity, camera, date range; results paginated
BR-021	The platform shall generate incident summary reports	Medium	Management requires visibility into incident volume and trends	Summary reports available by time period, severity, status, and camera/site

1.17.4 16.4 Evidence Management Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-022	The platform shall automatically capture evidence clips when security events occur	Critical	Manual evidence assembly is slow and creates chain-of-custody gaps	Evidence clips captured within 5 seconds of event trigger; clip includes pre- and post-event footage
BR-023	The platform shall generate content hash (SHA-256) for every evidence clip	High	Evidence integrity must be verifiable for legal and compliance use	SHA-256 hash generated at clip creation; hash stored with metadata
BR-024	The platform shall enforce role-based access control on evidence	Critical	Evidence access must be restricted to authorized personnel only	Unauthorized access attempts denied and logged
BR-025	The platform shall record chain-of-custody metadata for all evidence	Critical	Legal and compliance requirements demand complete evidence handling records	Every evidence access logged with user, timestamp, and action type
BR-026	The platform shall support configurable evidence retention policies	High	Retention requirements vary by regulation and organization policy	Retention policies configurable per site or incident type; expired evidence archived or deleted per policy
BR-027	The platform shall provide evidence retrieval within 10 seconds	High	Slow evidence retrieval delays investigations	Any evidence clip retrievable within 10 seconds of request
BR-028	The platform shall support evidence export with authorization	High	Evidence must be exportable for legal proceedings and external sharing	Export requires authorization; export action logged in audit trail

1.17.5 16.5 Dashboard Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-029	The platform shall provide a unified Security Operations Dashboard	Critical	Single pane of glass for all security operations activities	Dashboard displays live feeds, alerts, incidents, fleet status, and KPIs
BR-030	The platform shall render the dashboard within 2 seconds of login	High	Slow dashboard load impairs operator response capability	Dashboard fully loaded within 2 seconds on standard network connection
BR-031	The platform shall display real-time KPI metrics	High	Leadership requires visibility into security operational performance	KPIs including MTTD, alert volume, incident count, and SLA compliance displayed

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-032	The platform shall support customizable dashboard views per user role	Medium	Different roles require different operational views	Dashboard layout configurable per role; role-appropriate widgets displayed
BR-033	The platform shall display event timeline with filtering and search	High	Operators must correlate events temporally for investigation	Timeline view available; filterable by camera, event type, severity, time range

1.17.6 16.6 User Management Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-034	The platform shall enforce role-based access control across all modules	Critical	Security and compliance require controlled access to platform functions	Role-based permissions enforced; unauthorized access denied and logged
BR-035	The platform shall support predefined user roles (Operator, Supervisor, Administrator, System Admin)	High	Consistent permission model across enterprise deployments	Predefined roles available; permissions per role clearly defined
BR-036	The platform shall support custom role creation	Medium	Enterprise customers require flexible permission models	Custom roles definable with granular module and resource permissions
BR-037	The platform shall enforce authentication on all platform access	Critical	Unauthenticated access creates unacceptable security risk	All access points require valid authentication; unauthenticated requests rejected
BR-038	The platform shall enforce password policies (complexity, expiration, lockout)	High	Weak passwords are a primary attack vector	Configurable password policies; lockout after configurable failed attempt threshold

1.17.7 16.7 Administration Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-039	The platform shall provide administrative interface for system configuration	High	Administrators require centralized configuration management	Admin console accessible to authorized administrators; all configurations manageable

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-040	The platform shall support camera registration and configuration	Critical	Camera fleet must be manageable through the platform	Cameras registerable with site, zone, and configuration metadata
BR-041	The platform shall support rule configuration through administrative interface	High	Business rules must be configurable without code changes	Rule configuration UI available; rules saveable and activatable without restart
BR-042	The platform shall support user account management	High	User lifecycle must be manageable through the platform	User creation, modification, deactivation, and role assignment available

1.17.8 16.8 Reporting Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-043	The platform shall generate operational reports (alerts, incidents, SLA performance)	High	Management requires data-driven decision support	Operational reports available by configurable time period and dimensions
BR-044	The platform shall generate compliance reports (audit trails, access logs, evidence handling)	High	Regulatory compliance requires documented evidence of security controls	Compliance reports generated on demand; cover audit, access, and evidence dimensions
BR-045	The platform shall support report export in standard formats	Medium	Reports must be shareable with external auditors and stakeholders	Reports exportable in PDF and CSV formats
BR-046	The platform shall provide trend analysis on incident volume and severity	Medium	Trend data enables proactive security posture management	Trend reports available; visualized over configurable time periods

1.17.9 16.9 Compliance Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-047	The platform shall maintain immutable audit trails for all user and system actions	Critical	Regulatory compliance and legal defensibility require complete audit records	Every action logged with user, timestamp, action type, and affected resource; logs tamper-evident
BR-048	The platform shall enforce data retention policies per configurable policy	High	GDPR, CCPA, and other regulations require enforced retention periods	Retention policies configurable; expired data handled per policy

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-049	The platform shall support data access rights (right to access, right to deletion)	Medium	GDPR and CCPA require data subject access and deletion rights	Data access and deletion requests processable through administrative workflow
BR-050	The platform shall support configurable encryption at rest and in transit	High	Data protection regulations require encryption of sensitive data	AES-256 encryption at rest; TLS 1.3 encryption in transit; configurable per deployment

1.17.10 16.10 Notification Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-051	The platform shall deliver real-time notifications to the dashboard	Critical	Operators must receive alerts immediately in their primary workspace	Dashboard notifications delivered within 5 seconds of event generation
BR-052	The platform shall support email notifications for alerts and escalations	High	Off-hours and escalation scenarios require out-of-band notification	Email notifications sent within 60 seconds of trigger condition
BR-053	The platform shall support configurable notification rules per severity and role	High	Notification routing must align with organizational response models	Notification rules configurable; routing based on severity, role, and time of day
BR-054	The platform shall support webhook notifications for system integration	Medium	Enterprise customers require integration with existing notification infrastructure	Webhook notifications delivered to configured endpoints with retry logic

1.17.11 16.11 Audit Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-055	The platform shall record every user action in an immutable audit log	Critical	Every action must be traceable for compliance and security	Audit log captures user, timestamp, action, resource, and outcome for every action
BR-056	The platform shall record all system events (service startup, errors, configuration changes)	High	System events must be auditable for operational forensics	System events logged with timestamp, component, event type, and details

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-057	The platform shall support audit log query and filtering	High	Auditors and investigators must be able to search audit records	Audit logs searchable by user, date range, action type, and resource
BR-058	The platform shall enforce tamper-evident audit log storage	Critical	Audit integrity is a compliance requirement	Audit logs cryptographically signed or hashed; tampering detectable

1.17.12 16.12 Analytics Requirements

Req ID	Description	Priority	Business Justification	Acceptance Criteria
BR-059	The platform shall provide detection analytics by camera, zone, and time	High	Security teams need visibility into detection patterns	Analytics dashboard showing detection volume, distribution, and trends
BR-060	The platform shall provide camera utilization analytics (uptime, alert density)	Medium	Fleet management requires camera performance visibility	Camera utilization metrics available; uptime and alert density reported
BR-061	The platform shall provide operator performance analytics	Medium	Management requires visibility into operator productivity	Operator metrics including acknowledgment time, resolution time, and workload
BR-062	The platform shall support custom date range selection for analytics	Medium	Analysis requires flexibility in time period selection	Date range picker available across all analytics views

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1.18 17. Business Rules

1.18.1 17.1 Business Rules Register

Rule ID	Rule Description	Category	Enforcement
BR-R01	Only authenticated and authorized users may access the platform	Access Control	Mandatory
BR-R02	Every security alert must be acknowledged by an authorized operator	Alert Mgmt	Mandatory
BR-R03	Every security incident must have an assigned owner	Incident Mgmt	Mandatory
BR-R04	Critical-severity incidents must be escalated to supervisor within 5 minutes	Escalation	Mandatory
BR-R05	High-severity incidents must be acknowledged within 15 minutes	Escalation	Mandatory

Rule ID	Rule Description	Category	Enforcement
BR-R06	Evidence retention policies must be enforced regardless of system state	Compliance	Mandatory
BR-R07	Every user action must be recorded in the audit log	Audit	Mandatory
BR-R08	Evidence access must be restricted to users with appropriate role permissions	Evidence Mgmt	Mandatory
BR-R09	Rules defined in the Rule Engine cannot be bypassed by operators	Rule Mgmt	Mandatory
BR-R10	Incident SLA timers are non-deferrable and track continuous time	Incident Mgmt	Mandatory
BR-R11	Evidence content hashes must be verified on every access	Evidence Mgmt	Mandatory
BR-R12	Camera health status must be checked at least every 60 seconds	Fleet Mgmt	Mandatory
BR-R13	System availability must be maintained at 99.95% or higher	Operations	Mandatory
BR-R14	User accounts must be deactivated immediately upon role change or termination	Access Control	Mandatory
BR-R15	Audit logs must be tamper-evident and retained for a minimum of 12 months	Compliance	Mandatory

---

1.19 18. Business Constraints

1.19.1 18.1 Budget Constraints

Constraint	Description
Capital expenditure limits	Platform development must operate within approved engineering budget
Operational expenditure limits	Ongoing platform costs (compute, storage, networking) must be within operational budget
Licensing costs	Third-party dependencies (AI models, databases) must be cost-effective
Pilot deployment budget	Initial enterprise pilots must be delivered within allocated project budget

1.19.2 18.2 Timeline Constraints

Constraint	Description
MVP delivery timeline	MVP must be delivered within Phase 1 timeline (4 months)
Pilot customer commitments	Production deployments must meet committed customer timelines

Constraint	Description
Market window	Competitive positioning requires delivery before key market events
Regulatory deadlines	Compliance features must be available before customer audit cycles

1.19.3 18.3 Existing Infrastructure Constraints

Constraint	Description
Camera fleet diversity	Enterprise customers operate cameras from multiple vendors (Axis, Hanwha, Hikvision, etc.)
Network architecture	Camera networks are often segmented from application networks
IT governance	Deployments must comply with enterprise IT security policies
Integration requirements	Must integrate with existing SIEM, access control, and HR systems

1.19.4 18.4 Compliance Constraints

Constraint	Description
GDPR compliance	Video data handling, retention, access, and deletion must comply with GDPR
CCPA compliance	Data access and deletion rights must be supportable
HIPAA compliance	Healthcare deployments must meet HIPAA security requirements
SOC 2 compliance	Audit trails, access controls, and change management must meet SOC 2
IEC 62443 compliance	Industrial deployments must meet IEC 62443 cybersecurity requirements

1.19.5 18.5 Network Constraints

Constraint	Description
Bandwidth limitations	Camera stream ingestion must operate within available network bandwidth
Latency requirements	Real-time detection-to-alert latency must be < 5 seconds
Network segmentation	Platform must operate across segmented network architectures
Firewall restrictions	Deployment must work within standard enterprise firewall configurations

1.19.6 18.6 Resource Constraints

Constraint	Description
Engineering team size	Development team capacity limits feature velocity
Specialized skills	AI/ML engineering skills are scarce and must be allocated strategically
Operational staffing	Enterprise customers have limited security operations staff
Training capacity	User training must be achievable within deployment timelines

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1.20 19. Assumptions

Assumption ID	Assumption
A-01	Target customers have existing IP camera infrastructure with RTSP-capable cameras
A-02	Camera networks provide sufficient bandwidth for stream ingestion at required FPS
A-03	Customers have IT infrastructure capable of hosting Docker-based deployments
A-04	Security operations teams are available for platform training and adoption activities
A-05	Enterprise customers have standard network security controls (firewalls, VPNs)
A-06	AI detection models will achieve acceptable accuracy for core use cases in MVP
A-07	GPU resources are available for AI inference at scale
A-08	Enterprise customers have budget allocated for security intelligence platform investment
A-09	Regulatory requirements for target markets are well-defined and stable
A-10	Existing camera infrastructure will remain in place during platform adoption
A-11	Customers will provide adequate access to camera fleet for integration testing
A-12	Market demand for AI-powered physical security intelligence will continue to grow

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1.21 20. Business Risks

1.21.1 20.1 Enterprise Risk Register

Risk ID	Risk	Description	Probability	Impact	Owner	Mitigation	Status
BRK-01	AI model accuracy insufficient for production	Detection models may not achieve required accuracy levels, leading to excessive false positives or missed detections	Medium	High	AI/ML Team	Continuous model evaluation; human-in-the-loop validation; feedback loop from operators	Open
BRK-02	Camera compatibility issues across vendors	Diverse camera vendors may have inconsistent RTSP implementations, causing stream failures	High	Medium	Engineering	RTSP standard compliance; vendor-specific testing matrix; proactive vendor engagement	Open

Risk ID	Risk	Description	Probability	Impact	Owner	Mitigation	Status
BRK-03	Scalability limits under high camera counts	Platform may not meet performance targets at 10,000+ camera scale	Medium	High	Engineering	Performance testing at target scale; horizontal scaling design; load testing programs	Open
BRK-04	User adoption resistance from security teams	Operators may resist transition from familiar tools to new platform	Medium	Medium	Product Management	Phased rollout; operator-centric UX design; training programs; demonstrate quick wins	Open
BRK-05	Regulatory requirements vary by geography	GDPR, CCPA, and other regulations have conflicting requirements across markets	Medium	Medium	Compliance	Modular compliance layer; configurable data retention policies; legal review per market	Open
BRK-06	Dependency on upstream AI model availability	Open-source model availability or licensing may change	Low	High	AI/ML Team	Model versioning; offline inference capability; fallback detection modes; model ownership	Open
BRK-07	Competitive response from established VMS vendors	Legacy vendors may accelerate AI integration or acquire AI analytics companies	Medium	Medium	Product Management	Differentiate on integrated platform; speed of innovation; customer value delivery	Open
BRK-08	Data privacy backlash from AI-powered surveillance	Public or regulatory concern about AI-based video surveillance may create adoption barriers	Medium	High	Legal / Compliance	Privacy-by-design; transparent data handling; opt-out mechanisms; compliance-first approach	Open
BRK-09	Talent acquisition for AI/ML engineering	Difficulty hiring and retaining specialized AI/ML engineers	Medium	Medium	Engineering	Competitive compensation; open-source contributions; academic partnerships	Open
BRK-10	Pilot customer failure to achieve expected ROI	Pilot customers may not realize expected operational improvements	Low	High	Product Management	Success metric tracking; proactive customer success engagement; iterative improvement	Open

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1.22 21. Key Performance Indicators

1.22.1 21.1 Operational KPIs

KPI ID	KPI Name	Definition	Target	Measurement Method
KPI-01	Mean Time to Detect (MTTD)	Average time from event occurrence to system detection	< 5 seconds	System telemetry
KPI-02	Mean Time to Respond (MTTR)	Average time from alert generation to operator acknowledgment	< 30 seconds	Alert management reporting
KPI-03	Mean Time to Resolve	Average time from incident creation to resolution	40% reduction vs. baseline	Incident management reporting
KPI-04	False-Positive Rate	Percentage of alerts classified as false positive	< 10%	Alert classification analytics
KPI-05	Alert Acknowledgment Rate	Percentage of alerts acknowledged within SLA	> 95%	Alert management reporting

1.22.2 21.2 System KPIs

KPI ID	KPI Name	Definition	Target	Measurement Method
KPI-06	System Availability	Percentage of time system is operational and accessible	99.95%	Infrastructure monitoring
KPI-07	Camera Ingestion Uptime	Percentage of time camera streams are actively ingested	99.9% per stream	Health monitoring
KPI-08	Dashboard Load Time	Time from page request to fully rendered dashboard	< 2 seconds	Frontend performance monitoring
KPI-09	API Response Time (p95)	95th percentile API response time	< 200ms	API gateway metrics
KPI-10	Evidence Retrieval Time	Time to retrieve any evidence clip on request	< 10 seconds	Storage performance monitoring

1.22.3 21.3 Business KPIs

KPI ID	KPI Name	Definition	Target	Measurement Method
KPI-11	Operator Productivity	Incidents handled per operator per shift	30% improvement	Operational reporting

KPI ID	KPI Name	Definition	Target	Measurement Method
KPI-12	User Adoption Rate	Percentage of licensed operators actively using the platform weekly	> 80%	Session analytics
KPI-13	Customer Satisfaction	Net Promoter Score or equivalent customer satisfaction measure	> 40 NPS	Customer surveys
KPI-14	System Adoption Rate	Percentage of camera fleet actively monitored through the platform	> 90%	Fleet utilization analytics
KPI-15	Compliance Score	Audit finding resolution rate and compliance checklist completion	100%	Compliance reporting

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1.23 22. Success Criteria

1.23.1 22.1 Deployment Success

Criterion	Measurement	Target
MVP delivered on schedule	Phase 1 delivery within 4-month timeline	On-time
Pilot customers deployed	Production deployments at enterprise pilot customers	≥ 5 customers
Camera fleet coverage	Cameras actively monitored through the platform	> 90% of registered cameras
Core functionality operational	All Critical-priority business requirements implemented	100%

1.23.2 22.2 Operational Success

Criterion	Measurement	Target
MTTD achieved	Mean time to detect reduced from baseline	< 5 seconds
False-positive rate achieved	False-positive alerts as percentage of total alerts	< 10%
Incident resolution improved	Mean time to resolve reduced from baseline	40% reduction
Evidence retrieval	Time to retrieve any evidence clip	< 10 seconds
Dashboard performance	Time to full dashboard render	< 2 seconds

1.23.3 22.3 Customer Success

Criterion	Measurement	Target
Customer adoption	Operators actively using platform weekly	> 80%
Customer satisfaction	Net Promoter Score or equivalent	> 40 NPS
Customer retention	Pilot customers converting to annual contracts	> 80%
Reference customers	Customers willing to provide references	≥ 3 customers

1.23.4 22.4 Business Success

Criterion	Measurement	Target
Market validation	Enterprise customers across multiple verticals	≥ 3 verticals
Revenue trajectory	Recurring revenue run rate	Per business plan
Competitive win rate	Win rate against incumbent solutions	> 50%
Analyst recognition	Mention in relevant analyst reports	Within 18 months

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1.24 23. Business Acceptance Criteria

1.24.1 23.1 Business Acceptance Conditions

Before the business formally accepts the VigilantAI platform for production deployment, the following conditions must be met:

Acceptance ID	Condition	Verification Method
BAC-01	All Critical-priority business requirements (BR-001 through BR-062) are implemented	Requirements review and demonstration
BAC-02	MTTD target of < 5 seconds is achieved and sustained over a 30-day evaluation period	System telemetry and reporting
BAC-03	False-positive rate of < 10% is achieved and sustained over a 30-day evaluation period	Alert analytics
BAC-04	System availability of 99.95% is achieved during the pilot evaluation period	Infrastructure monitoring
BAC-05	Dashboard load time of < 2 seconds is consistently achieved	Performance testing
BAC-06	All evidence management requirements (chain-of-custody, access control, retention) are demonstrated	Compliance review
BAC-07	All audit trail requirements are met and verified by compliance review	Audit log review
BAC-08	Role-based access control is functional across all modules	Security testing
BAC-09	Security operations operators confirm usability and workflow alignment	User acceptance testing
BAC-10	Security management confirms operational visibility and KPI reporting meets requirements	Management review

Acceptance ID	Condition	Verification Method
BAC-11	Compliance officer confirms audit trail and evidence management compliance	Compliance sign-off
BAC-12	No Critical or High-severity defects remain unresolved	Defect management report
BAC-13	All pilot customer success metrics are tracked and reported	Customer success dashboard
BAC-14	Platform deployment and operation documented for ongoing operations	Operations runbook review

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1.25 24. Requirements Traceability Matrix

1.25.1 24.1 Traceability: Business Goal → Objective → Requirement → Metric

Business Goal	Business Objective	Business Requirement(s)	Success Metric
G-01: Deploy 3–5 pilot customers	OBJ-07: Deploy pilot customers	BR-001, BR-006, BR-013, BR-022, BR-029	≥ 5 active customer deployments
G-02: Reduce MTTD	OBJ-01: Reduce mean time to detect	BR-006, BR-009, BR-051	MTTD < 5 seconds
G-03: Viable alternative to legacy VMS	OBJ-02: Reduce false-positive rate	BR-008, BR-010, BR-030	False-positive rate < 10%
G-04: Scale to 20+ deployments	OBJ-06: Support enterprise camera fleet	BR-001, BR-004, BR-005, BR-040	50–10,000+ cameras per deployment
G-05: Expand AI model coverage	OBJ-08: Expand detection scenarios	BR-002, BR-003, BR-006	≥ 10 supported detection types
G-06: Build integration ecosystem	OBJ-09: Build integrations	BR-015, BR-054, BR-012	≥ 5 third-party integrations
G-07: Market recognition	OBJ-10: Achieve SaaS readiness	BR-029, BR-031, BR-043	Analyst recognition
G-08: 10,000+ cameras per customer	OBJ-06: Support enterprise camera fleet	BR-001, BR-004, BR-005	Demonstrated at ≥ 3 sites
G-09: Recurring revenue	OBJ-10: SaaS readiness	All platform requirements	Revenue target achieved
—	OBJ-03: Automate incident lifecycle	BR-013, BR-014, BR-015, BR-016, BR-017, BR-018, BR-019, BR-020	40% reduction in MTTR
—	OBJ-04: Eliminate manual evidence assembly	BR-022, BR-023, BR-024, BR-025, BR-026, BR-027, BR-028	Evidence retrieval < 10 seconds

Business Goal	Business Objective	Business Requirement(s)	Success Metric
—	OBJ-05: Deliver operational visibility	BR-029, BR-030, BR-031, BR-032, BR-033	Dashboard load < 2 seconds
—	—	BR-034, BR-035, BR-036, BR-037, BR-038	100% authenticated access
—	—	BR-047, BR-055, BR-056, BR-057, BR-058	Audit trail completeness
—	—	BR-048, BR-049, BR-050	Compliance score 100%
—	—	BR-051, BR-052, BR-053, BR-054	Notifications delivered within SLA

1.26 25. Glossary

Term	Definition
AI Detection Engine	The computer vision component that analyzes video frames to detect and classify objects
Alert	A notification generated when a security event meets configured rule criteria
Camera Fleet	The complete set of IP cameras managed through the VigilantAI platform
Camera Gateway	The ingestion service that connects to camera fleets via RTSP and manages stream lifecycle
Chain of Custody	The documented, tamper-evident record of evidence handling from capture to presentation
Event	A discrete occurrence generated by the event processor when rule conditions are met
Evidence	Video clips, snapshots, and metadata preserved for incident investigation
Evidence Integrity	The verifiable authenticity and completeness of evidence, maintained through content hashing
False Positive	An alert triggered by non-threatening activity (shadows, animals, lighting changes)
Incident	A security occurrence that is tracked from detection through resolution
Mean Time to Detect (MTTD)	Average elapsed time between event occurrence and system detection
Mean Time to Resolve (MTTR)	Average elapsed time from incident creation to resolution
Operator	A security operations team member who monitors alerts and manages incidents

Term	Definition
Restricted Zone	A configured area within a camera's field of view where unauthorized presence triggers an alert
Rule Engine	The component that evaluates detected events against configurable business rules
RTSP	Real Time Streaming Protocol — standard protocol for accessing live video streams from cameras
Security Event	A classified occurrence detected by the AI engine that may require operator attention
Security Operations Dashboard	The unified real-time console for monitoring, alert management, and incident workflows
SIEM	Security Information and Event Management — enterprise platforms for aggregating and analyzing security events
SLA	Service Level Agreement — defined response and resolution time commitments
VMS	Video Management System — traditional software for recording, storing, and viewing surveillance video

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1.27 26. Appendices

1.27.1 26.1 Abbreviations

Abbreviation	Full Form
ABAC	Attribute-Based Access Control
AES	Advanced Encryption Standard
AI	Artificial Intelligence
CCPA	California Consumer Privacy Act
CIO	Chief Information Officer
CTO	Chief Technology Officer
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IEC	International Electrotechnical Commission
IT	Information Technology
JWT	JSON Web Token
KPI	Key Performance Indicator
MTTD	Mean Time to Detect
MTTR	Mean Time to Resolve
MVP	Minimum Viable Product
NPS	Net Promoter Score
NVR	Network Video Recorder
RBAC	Role-Based Access Control
ROI	Return on Investment

Abbreviation	Full Form
RPO	Recovery Point Objective
RTO	Recovery Time Objective
RTSP	Real Time Streaming Protocol
SaaS	Software as a Service
SIEM	Security Information and Event Management
SLA	Service Level Agreement
SOC	Security Operations Center
SOC 2	Service Organization Control 2
TLS	Transport Layer Security
VMS	Video Management System

1.27.2 26.2 Reference Documents

Document	Description
VigilantAI Executive Summary (Document 01)	Product vision, architecture, and strategic overview
OWASP Top 10 (2021)	Web application security risks
GDPR — Video Surveillance Guidance	European data protection requirements for CCTV
HIPAA Security Rule	Healthcare facility security requirements
IEC 62443	Industrial cybersecurity standard
NIST SP 800-34	Contingency Planning Guide
SOC 2 Trust Services Criteria	Security, availability, processing integrity, confidentiality, and privacy controls

1.27.3 26.3 Related Documents

Document	Description
Technical Architecture Document	Detailed system architecture and component design
API Specification	REST API and WebSocket API reference documentation
Deployment Guide	Platform installation and configuration procedures
User Guide	End-user documentation for operators and administrators
Security Architecture Document	Detailed security controls and compliance mapping

1.27.4 26.4 Business References

Reference	Description
Global Video Surveillance Market Analysis (2024–2028)	Market sizing and growth projections
Enterprise Security Operations Benchmark	Industry benchmarks for security operations metrics
Physical Security ROI Study	ROI analysis for intelligent surveillance platforms

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End of Document